MGAIA Assignment 1: Procedural Content Generation

Leiden University

February 2026

1 Introduction

In this assignment, we want you to familiarize yourself with Procedural Content Generation (PCG). Minecraft is a host to many interesting AI challenges, such as settlement generation [Sal+18][1] (see Figure 1), or co-operative natural language interaction [Kis+21]. You will be generating a **House** in Minecraft.

Figure 1: Entry from the Generative Design in Minecraft Challenge (GDMC). This settlement was constructed algorithmically and not built manually.

[1] https://gendesignmc.engineering.nyu.edu/

2 Required Setup

1. Install the latest version of minecraft In the Official Minecraft Launcher. Before you go to the next step, press Play, otherwise it won't download. *Note that we of course discourage third-party launchers, such as TLauncher, that allow you to download the official client from the official minecraft website, without owning an account.*

2. Follow the steps for the Fabric installation for the GGD**M**C HTTP interface mod in the readme of: https://github.com/Niels-NTG/gdmc_http_interface/blob/master/README.md and Install the latest version of the mod (v1.8.1): https://github.com/Niels-NTG/gdmc_http_interface/releases/tag/v1.8.1. The release contains a *.jar file* that you need to install as a mod. Note that installing it simply means dragging it to the `%APPDATA%/.minecraft/mods/` folder (on Windows).

3. Install GD**P**C (the Python package you will be using). You need to clone https://github.com/avdstaaij/gdpc and install the package by running `pip install .` in the root of the cloned repository. Alternatively, you can install it using `pip install git+https://github.com/avdstaaij/gdpc` . **Do not use "pip install gdpc"**, as that will give you an outdated version.

4. Open Minecraft **with** the Fabric profile selected. Open a world, and you will see a chat message that the mod has opened an HTTP connection at the address localhost:9000. Set a build area[2], and try to run `examples/emerald_city.py`. If this works, you should now be ready.

5. Familiarize yourself with GDPC by going through the guides at `examples/tutorials`

3 Troubleshooting

Many problems (such as `ImportError: sys.meta_path is None, Python is likely shutting down`) can be resolved if you set the *buffering* argument of the `Editor` to "False".

4 Your Task

Your code will procedurally generate a building in the *build area*[2]. When testing your submission, we will use a build area of 100x100 blocks. Of course, you don't know where we will test it, so your building should smoothly integrate into the existing world. Do not just clear all blocks in a bounding box, but make sure your algorithm attempts to scan the selected environment, and places its structures in a fitting way. This is the difficulty and essence of Procedural Content Generation! Adaptability to the infinite possibilities that RNGesus might throw at you.

Moreover, you will be writing a scientific report. There you will make the intentions behind your architectural design clear, as well as highlight inspirations and applied algorithms to achieve your goal.

Your generated structure should include *at minimum* the things below, but more can only be better.

[2] In Minecraft, press `t` and type `/buildarea set [x0] [y0] [z0] [x1] [y1] [z1]` (https://minecraft.wiki/w/Coordinates#Commands). The easiest way to use this command is by writing the following `/buildarea set ~ ~ ~ ~100 ~100 ~100` which will make a 100x100x100 buildarea based on where your character is currently standing

- A house of a specific architectural style (be sure to give an example in your report)
- Interior & Exterior decoration (so furniture and garden)
- Appropriate adaptation to the terrain in the build area

We expect you to attempt to implement these things in a way that the world is *believable*. Believable in the sense that, e.g., a house should not spawn directly in the middle of an ocean where it would not be reachable. And ideally, you do not "heavily" alter the area where it should be placed, but of course if you are in the middle of a jungle and there is no continuous free space to put your structure due to too many trees, you are allowed to make free space.

While placing your building in a believable way is challenging we do want you to focus on creativity. The magic of PCG is that every time you run the code you generate something slightly different. That is why 40% of your grade for this assignment stems from the variation your code will generate. Think of varying building layouts and sizes, varying interior and exterior placement, etc. The keyword is **Procedural Content Generation**, not place a given structure at point X,Y,Z.

5 Submission

Make sure to nicely document everything that you do in the report. Your final submission consists of:

- Source code for a python program that uses the GD**P**C Python package and integrates a House believably into the world.
  - Your code should be runnable with a single command (e.g. "python mypcg.py") and not require any other python libraries besides gdpc, what it includes and at most numpy/scipy/matplotlib.
- A self-contained scientific pdf report with figures, **references**, etc. The page count we expect of you might vary depending on your layout (4-8 pages would be reasonable). This report contains:
  - An explanation of the techniques you applied, e.g.
    - How do you find an area to place your building? Include plots that visualize your terrain evaluation
    - What is variable / random in the building process?
  - How you addressed *believablity*.
  - **Multiple** example figures showing the *variation* as well as *adaptability* to different environments.
  - Overall conclusions.

If you have any questions about this assignment, please visit our lab sessions on Thursdays where we can help you out. In case you cannot make it in person, you can post questions about the contents of the course on the Discord discussion forum, where other students can also read and reply to your questions.

The deadline for this assignment is **05.03.2026 at 23:59**. -0.5 points will be deducted per 24 hours late submission.

6 Grading Criteria

Your grade for the assignment will be made up of:

- 35% Believability & Adaptability (e.g. Architectural Style, Terrain adaptation, plots of terrain evaluation)
- 40% Randomness/Variation of Procedurally Generated Content (e.g. varying shape, size and orientation, interior & exterior decoration)
- 25% Report (e.g. Structure, Explanation, Example images, Conclusions/Discussions, Scientific References / Related work, Writing Style, etc.)

7 References

References

[Kis+21] Julia Kiseleva et al. "Neurips 2021 competition iglu: Interactive grounded language understanding in a collaborative environment". In: *arXiv preprint arXiv:2110.06536* (2021).

[Sal+18] Christoph Salge et al. "Generative design in minecraft (GDMC) settlement generation competition". In: *Proceedings of the 13th International Conference on the Foundations of Digital Games*. 2018, pp. 1–10.

8 Examples from previous year

Note that these images are solely meant as inspiration and do not serve as a solution you should be attempting to copy exactly.

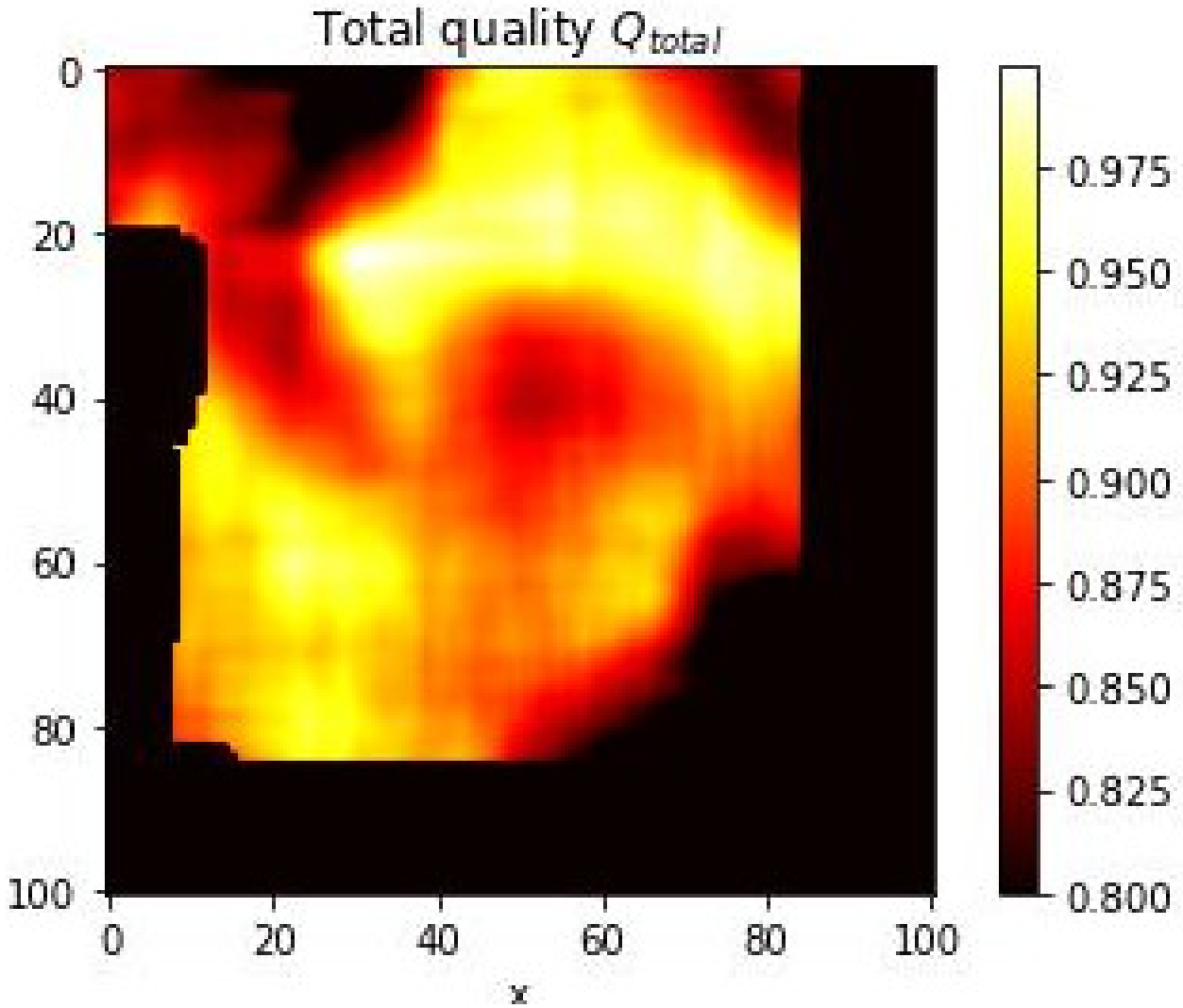

Figure 2: An example plot in which students visualize the final evaluated suitability of the building area. This was preceded by a few other individual plots that show individual features that have been combined in this plot.

Figure 3: An example in which students procedurally generate a tree including houses as leaves

Figure 4: An example in which the student highlights the variety of building styles and sizes